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Evaluation of Autologous Fascia Implantation With Controlled Release of Fibroblast Growth Factor for Recurrent Laryngeal Nerve Paralysis Due to Long-term Denervation

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Abstract

Objectives: Paralyzed tissue due to long-term denervation is resistant to many treatments because it induces irreversible histological changes and disorders of deglutition or phonation. We sought to determine the effect of autologous transplantation of fascia into the vocal fold (ATFV) with controlled release of basic fibroblast growth factor (bFGF) on long-term unilateral vocal fold paralysis (UVFP).

Methods: Unilateral recurrent laryngeal nerve (RLN) section was performed on 20 rats. Five rats were implanted with autologous fascia only (fascia group), and 10 rats were implanted with autologous fascia and a gelatin hydrogel sheet with 1 µg (1 µg bFGF + fascia group) or 0.1 µg (0.1 µg bFGF + fascia group) of bFGF 4 months after RLN section. We evaluated the normalized glottal gap and laryngeal volume and histological changes 3 months after implantation.

Results: The normalized glottal gap was significantly reduced in the 3 fascia implantation groups. Normalized laryngeal volume, fat volume, and lateral thyroarytenoid muscle volume were significantly increased in the 2 fascia implantation with bFGF groups.

Conclusions: The ATFV with controlled release of bFGF repaired the glottal gap and laryngeal volume after RLN section and may reduce the occurrence of aspiration and hoarseness. We speculate that this treatment improves laryngeal function in long-term RLN denervation.

Keywords

larynx, basic fibroblast growth factor, vocal fold paralysis, tissue regeneration, laryngeal functional disorders

Introduction

Paralysis of recurrent laryngeal nerve (RLN) may result from various causes, including brain infarction, trauma, tumor, and iatrogenic injury. Disorders of RLN result in vocal fold paralysis due to the inability to transmit neural information to the intralaryngeal muscle. Moreover, as the duration of nerve paralysis increases, the innervated muscle atrophies, capillary vessels decrease or disappear, and collagen formation increases in the paralyzed tissue.[1-3] It is difficult to regenerate these altered tissues, even if neural transmission recovers. The treatment of unilateral vocal fold paralysis (UVFP) is not usually prioritized when treating the primary disease because surgical treatment of UVFP intervenes in laryngeal tissue atrophy.

Autologous transplantation of fascia into the vocal fold (ATFV), a treatment for UVFP, has been shown to improve the glottal gap.[4] It is possible that the implanted fascia provides a scaffold and cells and surgical stress activates regulatory factors in the patient. This is consistent with Langer and Vacanti's[5] theory that these 3 components are necessary for successful tissue regeneration. We hypothesized that ATFV with growth factors would improve the outcome of UVFP.

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This study aimed evaluate the effects on tissue formation of long-term denervation treated by ATFV with basic fibroblast growth factor (bFGF) using an animal model of UVFP.

Materials and Methods

This study used 9- to 12-week-old Sprague-Dawley (SD) male rats (weight: 485.5 g ± 92.6 g). Since the average life span of these rats is 36 months, we considered long-term RLN paralysis to be more than 4 months after nerve section.[6,7] Autologous fascia implantation into the paralyzed glottal tissue was performed 4 months after RLN section, and the larynxes were evaluated 3 months after implantation. These protocols were approved by the Kitasato Ethics Committee (Approval No. 2012-71, 2013-087). Animal care was conducted according to the Kitasato University of Medicine animal care manual.

The first operation involved preparing the RLN paralysis model. The left RLN and accompanying vessel, which was approximately 5 or 4 mm long, were removed from each anesthetized rat. The second operation involved implantation into the paralyzed larynx 4 months after RLN section. Specifically, autologous thoracolumbar fascia was harvested and squeezed tightly for dehydration and then shaped into a 1 × 1 mm square. Simultaneously, a same sized gelatin hydrogel sheet (gelatin sheet) was prepared. This was followed by implantation of the fascia and gelatin sheet with or without bFGF. A midline incision was made in the anterior neck. The strap muscle was then cleaved at its attachment to the hyoid bone, exposing thyroid and cricoid cartilage. The left cricothyroid muscle was cleaved at the inferior edge of the thyroid cartilage, and the thyroid lamina was vertically sectioned. An implantation bed was secured under the thyroid lamina in order to ablate around the connective tissue. The implanted materials were planted onto this bed, which was located on the lateral side of the thyroarytenoid (TA) muscle and lateral cricothyroid muscle[8] (Figure 1).

Gelatin sheets were prepared from an aqueous solution of glutaraldehyde and gelatin, as previously reported.[9,10] With the water content of the hydrogel sheet at 96 wt%, bFGF was released for 2 weeks. The gelatin hydrogel was trimmed into 1 × 1 mm squares, which were then soaked in an aqueous solution containing 0.1 or 1 μg bFGF at room temperature 3 hours before use.[8]

Rats were divided into the following 4 groups: a group implanted with fascia and gelatin sheet with 1 μg bFGF (1 μg bFGF + fascia group, n = 5), a group implanted with fascia and gelatin sheet with 0.1 μg bFGF (0.1 μg bFGF + fascia group, n = 5), a group implanted with fascia and gelatin sheet without bFGF (fascia group, n = 5), and a group without implantation (RLN section group, n = 5).

Rats were anesthetized with ether followed by pentobarbital (30 mg/kg at i.p.). In cases of insufficient anesthesia, we administered additional pentobarbital (0.2 ml at i.p.). We attempted to minimize stress during all study procedures. Each rat was euthanized by CO_2 inhalation. Before euthanasia, vocal fold movement was observed by oral insertion of a rigid endoscope after opening the mouth with a micro laryngoscope. All rats exhibited loss of left arytenoid cartilage movement on the side of RLN section (Figure 2).

After euthanasia, the larynx was removed, fixed with 10% formalin, dehydrated with graded concentrations of ethanol, and embedded in paraffin. Serial coronal paraffin sections (3 μm thick) were made at 100 μm intervals,

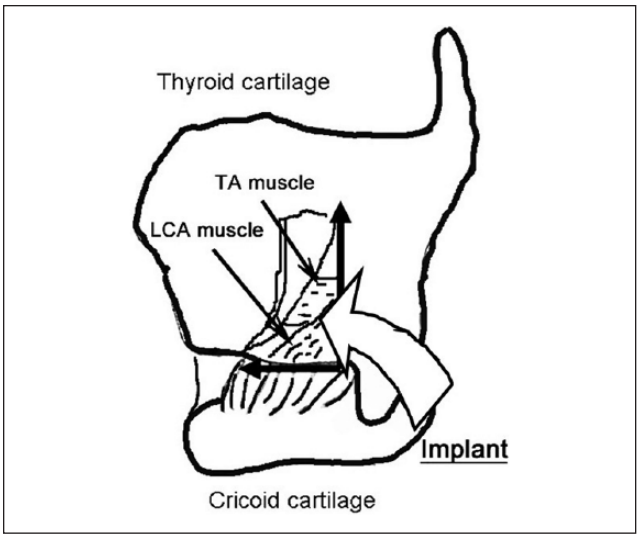


Figure 1. Representative schematic of implantation. The implanted materials were planted onto this implanted bed, which was located on the lateral side of the thyroarytenoid muscle (TA) muscle and lateral cricothyroid (LCA) muscle.

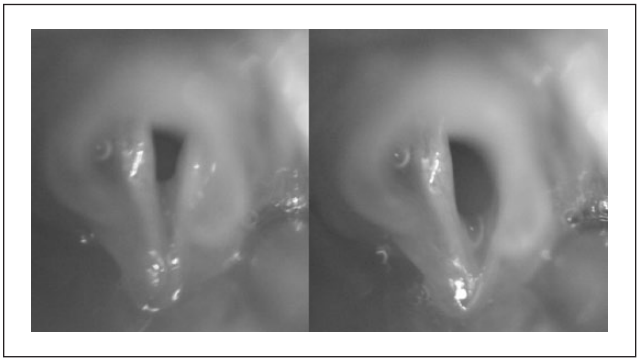


Figure 2. Laryngeal image after recurrent laryngeal nerve (RLN) transection during spontaneous respiration. Left image: adductive phase; right image: abductive phase. The left side of each image corresponds to the paralyzed side and the right side the unparalyzed side. Vocal fold at rat was very small because we recognized arytenoid cartilage. Left arytenoid cartilage was impassive at both phases. Right arytenoid abducted at inspiration phase and adducted a little at expiratory phase.

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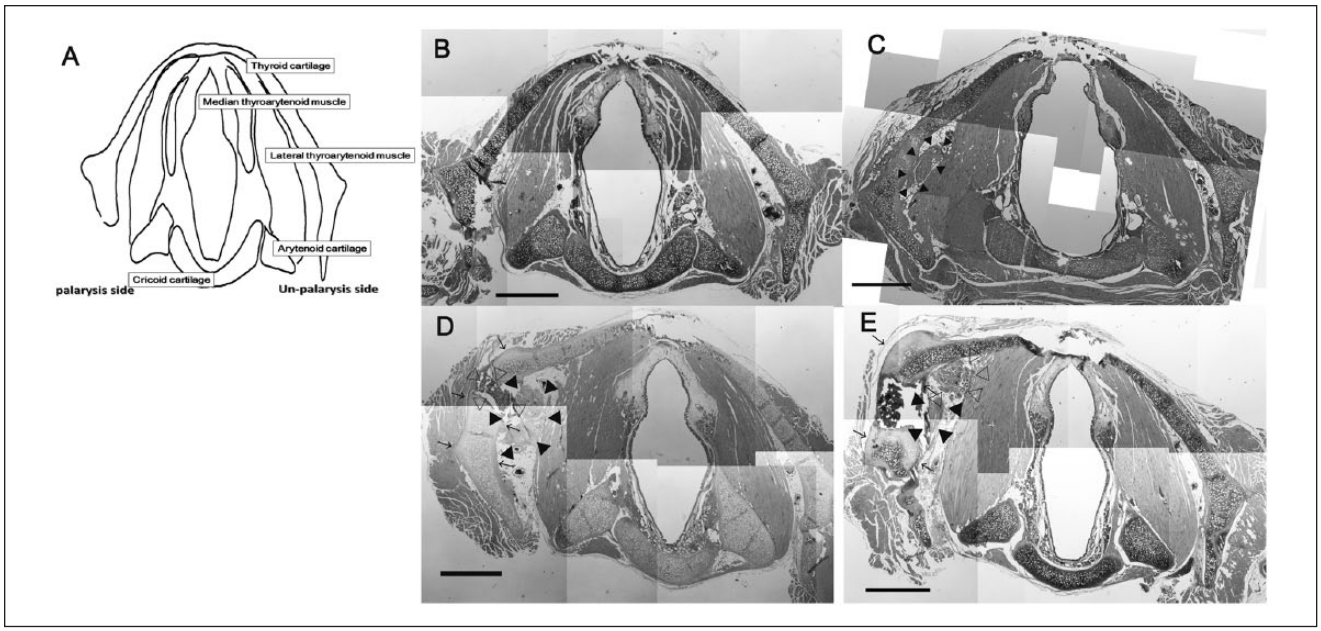


Figure 3. Representative images of the larynx for the 4 groups. (A) Explanation of the larynx image. (B) Recurrent laryngeal nerve group. (C) Fascia group. (D) 0.1 μg basic fibroblast growth factor (bFGF) + fascia group. (E) 1 μg bFGF + fascia group. Implanted fascia is shown within the black triangle, concentration of infiltrating cells within the triangle, and hypertrophy of thyroid cartilage is indicated by the arrow. The left side of each image corresponds to the paralyzed side and the right side the unparalyzed side. Scale: 1 mm.

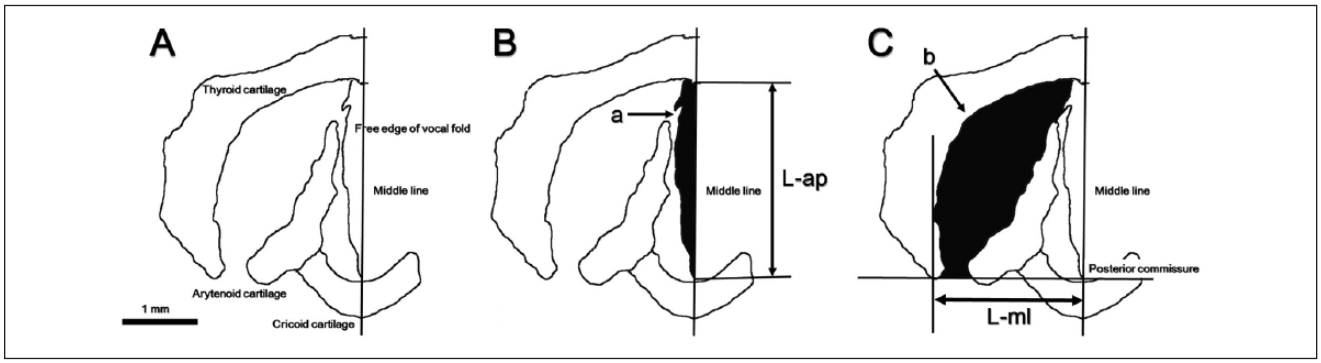


Figure 4. (A) Explanation of schematic. Schematic of (B) normalized glottal gap and (C) laryngeal volume. A schematic of superimposed slides from a rat larynx shows the minimum area of the glottal gap (a), anterior to posterior length (L-ap), minimum volume of laryngeal space (b), and maximum edge of the laryngeal volume to midline length (L-ml). Scale is 1 mm.

stained with hematoxylin and eosin, then viewed and photographed at low power with a digital camera (Canon, Tokyo, Japan) and light microscope (Nikon, Tokyo, Japan). Stained sections for each group were evaluated for histological alterations. Implanted fascia, adipose tissue, and intralaryngeal muscle, but not hypertrophy of thyroid cartilage or a concentration of infiltrating cells, were observed (Figure 3). Images were saved, and a whole laryngeal image was constructed using Adobe Photoshop Elements (Adobe, San Jose, California, USA). We revised each laryngeal image to use the saved micrometer in same requirement. The glottal gap and larynx volume were measured for each rat to determine the effect of these treatments.

Each laryngeal image was bisected down the midline, with only the left, treated side used for analysis. The edges of the cartilage and glottal lumen were traced in these half images. The traced images were superimposed in 2 places, 1 at the intersection between the midline and posterior surface of the laryngeal cavity and the other on the midline (Figure 4A). These superimposed images were imported into Scion Image (Scion Corporation, Torrance, California, USA) to measure the minimum area of the glottal space (a) and length from the anterior to posterior commissure at the

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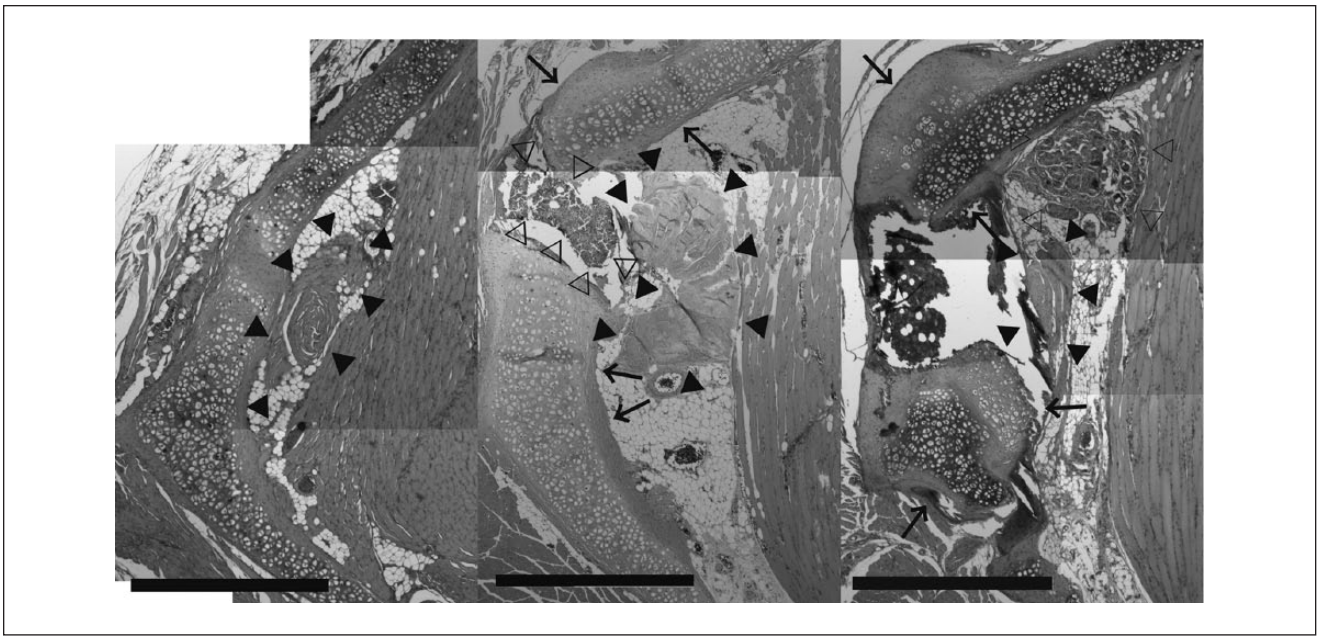


Figure 5. Representative high magnification image of implanted fascia and thyroid cartilage at treated side. Left image: fascia group; middle image: 0.1 μg basic fibroblast growth factor (bFGF) + fascia group; right image: 1 μg bFGF + fascia group. Hypertrophy of the thyroid cartilage and infiltrating cells were observed in the 2 bFGF + fascia groups. Implanted fascia is shown within the black triangle, concentration of infiltrating cells within the triangle, and hypertrophy of thyroid cartilage is indicated by the arrow. Scale: 1 mm.

midline (L-ap). The determined glottal gap closure was normalized to L-ap, and the rate of glottal gap closure was calculated using the following formula: normalized glottal gap area (units) = a / (L-ap)^2 × 100 units (Figure 4B).[8,11]

Evaluation of larynx volume used superimposed images only at the arytenoid cartilage level. Laryngeal tissue was very thin at the cricoid cartilage level. We speculated that the effect of this treatment was related to the laryngeal volume around the arytenoid cartilage because this was at the position of minimal glottal space in the upper airway. The posterior end of this laryngeal area provided the posterior position of the posterior commissure. These superimposed images were imported into Scion Image to measure the minimum volume of the laryngeal area (b) and the maximum length of the minimum laryngeal volume image from the midline (L-ml). The determined laryngeal volume was normalized to L-ml, and the rate of laryngeal volume change was calculated using the following formula: normalized laryngeal volume (units) = b / (L-ml)^2 × 100 units (Figure 4C).

Measurements of the implanted fascia, fat tissue, and lateral thyroarytenoid (lTA) muscle volume were performed for each stained slice using Scion Image. The volume of each parameter was determined by the formula: volume = sum of whole volume × 100 μm. Fat and lTA muscle were compared to the volume of the untreated side (%).

Each parameter was compared between groups using the Kruskal-Wallis rank test. Significant differences were analyzed with the Bonferroni-Dunn test for multiple comparisons. StatView for Windows (Ver 5.0, SAS Instrument, SAS, Cary, North Carolina, USA) was used for statistical analyses, and $P < .05$ was considered statistically significant.

Results

Histological examination revealed hypertrophy of the thyroid cartilage in cases of autologous fascia implantation in the bFGF groups, with the exception of 1 rat in the 0.1 μg bFGF + fascia group. The fascia implanted group and RLN section group did not exhibit changes in thyroid cartilage. Infiltrating cells were observed in the 0.1 μg bFGF + fascia group and the 1 μg bFGF + fascia group, whereas no such cells were observed in the fascia implanted group and RLN section group (Figure 5).

The normalized glottal gap area was significantly different between groups using the Kruskal-Wallis rank test ($P = .0022$). The areas in the treatment groups (the fascia implantation with bFGF groups and the fascia group) were significantly smaller than in the RLN section group ($P < .001$). The area in the 1 μg bFGF + fascia group was significantly smaller than in the fascia group ($P = .0047$; Figure 6).

The normalized laryngeal volume was also significantly different between groups ($P = .0024$). The volume in the 1 μg bFGF + fascia group was larger than in the RLN section and fascia groups ($P < .001$ and $P = .0042$, respectively). Similarly,

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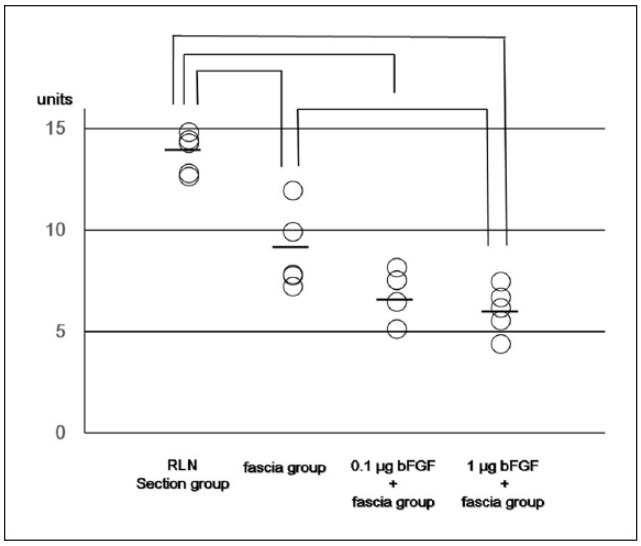


Figure 6. Normalized glottal gap. The fascia implantation groups had significantly smaller gap areas compared to the RLN section group (13.8 ± 1.0 units). The area in the fascia implantation with 1 µg bFGF (6.1 ± 1.2 units) was significantly smaller than with fascia implantation only (8.9 ± 2.0 units). bFGF, basic fibroblast growth factor; RLN, recurrent laryngeal nerve.

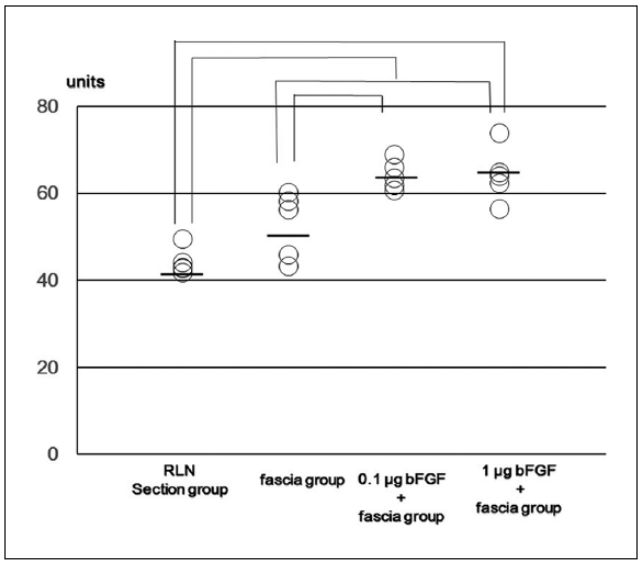


Figure 7. Normalized laryngeal volume. Volumes in the fascia implantation with bFGF groups (0.1 µg bFGF + fascia group, 64.2 ± 3.3 units; 1 µg bFGF + fascia group, 64.3 ± 6.3 units) were significantly different compared to the RLN section group (44.2 ± 3.1 units) and the fascia group (52.8 ± 7.7 units). bFGF, basic fibroblast growth factor; RLN, recurrent laryngeal nerve.

the volume in the 0.1 µg bFGF + fascia group was larger than in the RLN section and fascia groups ($P < .001$ and $P = .0044$, respectively). There was no significant difference between the 2 fascia implantation with bFGF groups (Figure 7).

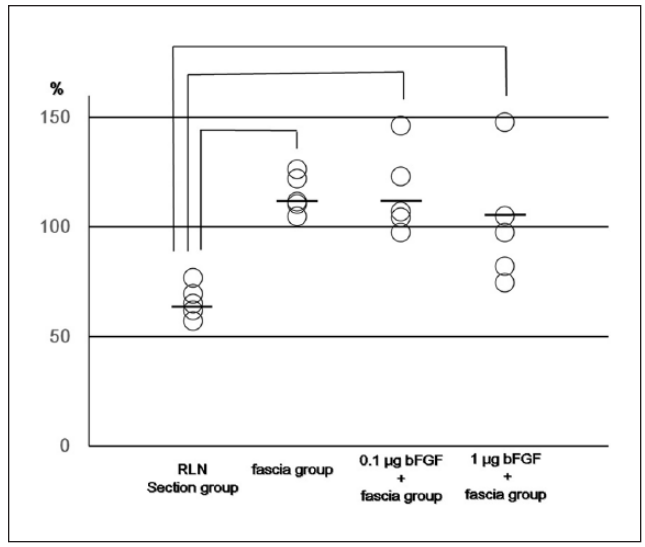


Figure 8. Percentage of lateral thyroarytenoid muscle volume. The fascia implantation groups (fascia group, 115.0 ± 8.8%; 0.1 µg bFGF + fascia group, 115.6 ± 19.5%; 1 µg bFGF + fascia group, 101.4 ± 28.6%) had significantly higher percentages of lTA muscle volume compared to the RLN section group (66.0 ± 7.5%). bFGF, basic fibroblast growth factor; lTA, lateral thyroarytenoid; RLN, recurrent laryngeal nerve.

The volume of implanted fascia was significantly different between groups ($P = .042$). Specifically, the 0.1 µg bFGF + fascia group had a significantly larger volume than the fascia group ($P = .0075$).

The percentage of fat volume was significantly different between groups ($P = .0031$). Fat volumes in the 0.1 µg bFGF + fascia group and the 1 µg bFGF + fascia group were significantly larger than in the RLN section group ($P = .0035$ and $P = .0061$, respectively).

The percentage of lTA muscle volume was also significantly different between groups ($P = .0095$). Percentages in the 3 fascia implantation groups (0.1 µg bFGF + fascia, 1 µg bFGF + fascia, and fascia groups) were significantly larger than in the RLN section group ($P = .0006$, $P = .0074$, and $P = .0006$, respectively; Figure 8).

Discussion

Unilateral vocal fold paralysis treatment will likely continue to be delayed because treatment of the primary disease is prioritized. This delay results in paralysis and changes in laryngeal tissue, including intralaryngeal muscle atrophy, destroys the nerve-vessel system, and increases fibrosis.[12-14] Treating long-term, paralyzed laryngeal tissue is difficult. Prolonged RLN denervation induces bowing of the vocal fold by increasing the glottal gap. The size of the glottal gap during closure is correlated with UVFP symptoms. A large glottal gap causes strong hoarseness and increases the risk

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of aspiration.[15,16] Additionally, expansion of the glottal gap reduces laryngeal area volume. This study examined the effects of autologous fascia implantation with bFGF on long-term denervation of laryngeal tissue in UVFP. Histological examination showed that glottal gap closure significantly improved with the treatment. Moreover, fat volume and lateral thyroarytenoid muscle volume increased compared to the recurrent laryngeal section control group. The diminished glottal gap and increased laryngeal volume suggest that this method is a valid treatment for UVFP.

Early intervention for UVFP is more effective than late intervention.[17-19] The duration of denervation influences the degree of intralaryngeal muscle atrophy and connective tissue fibrosis. This altered tissue is resistant to many treatments. Despite this, Miyamaru et al[6] found that nerve-muscle pedicle implantation was effective for long-term denervation of intralaryngeal muscle. Similarly, we found that fascia implantation with bFGF treatment significantly decreased the glottal gap and increased laryngeal volume compared to other groups, even with the delayed timing of intervention. We speculate that these treatments can improve tissue that has been denervated for a long period of time.

The intralaryngeal muscles contribute to laryngeal volume. The lTA muscle, an intralaryngeal muscle, decreases in volume after RLN paralysis. Kumai et al[12] reported that the TA muscle atrophied by approximately 53% 10 weeks after RLN section. Atrophy of the lTA muscle could be attributed to incomplete glottal closure and result in aspiration and hoarseness. Nerve-muscle pedicle implantation into long-term denervated tissue prevents lTA muscle volume decrease and atrophy, suggesting the possibility of developing an effective treatment for atrophy resulting from long-term denervation.[6] In this study, lTA muscle volumes in the autologous fascia implantation groups were significantly larger than in the RLN section group. Nishiyama et al[4] reported that ATFV for UVFP improved voice function. The lack of lTA muscle atrophy may have maintained the laryngeal volume on the paralyzed side. In contrast, the present study found no significant difference in the increase in lTA muscle volume between fascia implantation with or without bFGF. We speculate that bFGF was effective for ATFV because bFGF has been reported to induce satellite cell propagation[20] and vascularization in the skeletal muscle.[21] Moreover, muscle regeneration is related to vessel increases around the damaged muscle,[22] which led to the idea that addition of bFGF to autologous fascia implantation would be effective.

Controlled release of bFGF from implanted scaffolds has been reported to promote migration, proliferation, and maturation of adipose precursor cells, leading to an increase in adipose tissue volume.[23] Tamura et al[24] reported that fat implantation with controlled release of bFGF prevented the resorption of implanted fat in dogs with vocal fold paralysis. In the present study, fascia implantation with controlled release of bFGF significantly increased adipose tissue volume and induced angiogenesis. Isogai et al[25] reported that combined chondrocyte-copolymer implantation with slow release of bFGF induced neovascularization around implanted material. In another study, Iwakura et al[26] found that peristernal blood flow improved at the devascularized sternum upon controlled release of bFGF. These angiogenic effects may have promoted regeneration of damaged tissue. While these prior studies studied angiogenic effects at 4 weeks and 10 weeks after implantation, we studied histological changes at 12 weeks after implantation, given the lack of angiogenic changes at the implantation site at earlier time points.

Vocal fold injection is an effective treatment for UVFP.[27] The implantation of autologous material helps avoid a host reaction; however, these materials do not preserve volume due to resorbance over time.[28] We speculate that regeneration of the paralyzed laryngeal tissue could prevent a reduction in laryngeal volume due to resorption. Langer and Vacanti[5] proposed that regeneration was induced by 3 factors (cells, scaffold, and growth factors), and ATFV with bFGF provided these factors in the implanted area. ATFV with 1 μg bFGF decreased the glottal gap even though the implanted material resorbed at the same rate as the fascia implanted group. Thus, the increased laryngeal volume on the treated side was due to a factor in addition to the implanted material, most likely bFGF.

Basic fibroblast growth factor is a growth factor with a short half-life, so its controlled release is more effective in certain situations.[21] Our results may be further improved by examining the adequate dose of bFGF, which must be optimized for each species. Ischemic free-flap implantation increases survival rate in a dose-dependent manner.[29] For example, in 1 study, adipose regeneration induced by the controlled release of 1 μg bFGF was greater than that induced by other dosages.[23] Furthermore, cruciate ligament reconstruction in rats is not significantly different using polycaprolactone with or without the controlled release of bFGF at 0.1 μg.[30] High-dose bFGF produces more granulation tissue. The larynx is located at the smallest part of the airway, thus bFGF release needed to be carefully controlled. In the present study, ATFV with controlled release of 0.1 μg and 1 μg bFGF significantly decreased the glottal gap and increased laryngeal volume, demonstrating that both doses were adequate for use in combination with fascia implantation.

In our histological analysis, we found hypertrophy of the thyroid cartilage, a concentration of infiltrating cells, and an increase in adipose tissue occupying the increased volume of the implanted area for ATFV with bFGF compared to the fascia group. These alterations may maintain the volume of the laryngeal space after long-term denervation and suggest that bFGF improves the effects of ATFV for vocal fold paralysis.

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A limitation of this study is that the long-term effects of the implanted material on adipose tissue and atrophic muscle were not assessed. Thus, long-term assessments are warranted in the future.

Conclusions

This study revealed that ATFV with bFGF diminished the glottal gap in UVFP by increasing laryngeal volume. Histological examination revealed that adipose tissue and lTA muscle volume increased. These results suggest that ATFV with controlled release of bFGF improves laryngeal function lost due to long-term denervation in UVFP.

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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References

1. Sunderland S, Ray LJ. Denervation changes in mammalian striated muscle. *J Neurol Neurosurg Psychial*. 1950;13(3):159-177.
2. Borisov AB, Huang SK, Carlson BM. Remodeling of the vascular bed and progressive loss of capillaries in denerveted skeletal muscle. *Anat Record*. 2000;258(3):292-304.
3. Borisov AB, Dedkov EI, Carlson BM. Interrelations of myogenic response, progressive atropy of muscle fiber, and cell death in denervated skeletal muscle. *Anat Record*. 2001;264:203-218.
4. Nishiyama K, Nagai H, Masaki T, et al. Long-term result of the new endoscopic vocal fold medialization surgical technique for laryngeal palsy. *Laryngoscope*. 2006;116:231-234.
5. Langer R, Vacanti JP. Tissue engineering. *Science*. 1993;260:920-926.
6. Miyamaru S, Kumai Y, Ito T, Sanuki T, Yumoto E. Nerve-muscle pedicle implantation facilitate re-inervation of long-term denervated thyroarytenoid muscle in rats. *Acta otolaryngologica*. 2009;129:1486-1492.
7. Nishimoto K, Kumai Y, Sanuki T, Minoda R, Yumoto E. The impact of nimodipine administration combined with nerve-muscle pedicle implantation on long-term denervated rat thyroarytenoid muscle. *Laryngoscope*. 2012;123:952-959.
8. Nagai H, Nishiyama K, Seino Y, et al. Fascia implantation with fibroblast growth factor on vocal fold paralysis. *Am J Otolaryngol*. 2013;34:331-336.
9. Tabata Y, Ikada Y. Vascularization effect of basic fibroblast growth factor released from gelatin hydrogels with different biodegradabilities. *Biomaterials*. 1999;20:2169-2175.
10. Tabata Y, Miyao M, Inamoto T, et al. De novo formation of adipose tissue by controlled release of basic fibroblast growth factor. *Tissue Eng*. 2000;6:279-289.
11. Ohno T, Hirano S, Kanemaru S, et al. Drug delivery system of hepatocyte growth factor for the treatment of vocal fold scarring in a canine model. *Ann Otol Rhinol Laryngol*. 2007;116:762-769.
12. Kumai Y, Ito T, Matsukawa A, Yumoto E. Effect of denervation on neuromuscular junctions in the thyroarytenoid muscle. *Laryngoscope*. 2005;115:1869-1872.
13. Motoyoshi K, Hyodo M, Yamagata T, Gyo K. Restoring vocal fold movement after transection and immediate suturing of the recurrent laryngeal nerve with local application of basic fibroblast growth factor: an experimental study in the rat. *Laryngoscope*. 2004;114:1247-1252.
14. Woodson GE. Spontaneous laryngeal reinnervation after recurrent laryngeal or vargus nerve injury. *Ann Otol Rhinol Laryngol*. 2007;116:57-65.
15. Bielamowicz S, Kapoor R, Schwartz J, Stager SV. Relationship among glottal area, static supraglottic compression, and laryngeal function studies in unilateral vocal fold paresis and paralysis. *J Voice*. 2004;18:138-145.
16. Inagi K, Khidr AA, Ford CN, Bless DM, Heisey DM. Correlation between vocal functions and glottal measurements in patients with unilateral vocal fold paralysis. *Laryngoscope*. 1997;107:782-791.
17. Friedman AD, Burns JA, Heaton J, Zeitels SM. Early versus late injection medialization for unilateral vocal cord paralysis. *Laryngoscope*. 2010;120:2042-2046.
18. Li M, Chen S, Wang W, et al. Effect of duration of denervation on outcomes of ansa-recurrent laryngeal nerve reinnervation. *Laryngoscope*. 2014;124:1900-1905.
19. D'Alayri L, Galla S, Rigante M, Antonelli O, Buldrini S, Marchese M. Role of early voice therapy in patients affected by unilateral vocal fold paralysis. *J Laryngol Otol*. 2008;122:936-941.
20. Kastner S, Elias MC, Rivera AJ, Yablonka-Reuveni Z. Gene expression patterns of fibroblast growth factors and their receptors during myogenesis of rat satellite cells. *J Histochemical Soc*. 2000;48:1079-1096.
21. Rifkin DB, Moscatelli D. Recent developments in the cell biology of basic fibroblast growth factor. *J Cell Biology*. 1989;109:1-6.
22. Efthimiadou A, Nikolettos NK, Lambropoulou M, Papadopoulos N, Kontolen E. Angiogenic effect of intramuscular administration of basic fibroblast growth factor in atrophied muscles: an experimental study in the rat. *Br J Sports Med*. 2006;40:355-358.
23. Kimura Y, Ozeki M, Inamoto T, Tabata Y. Adipose tissue engineering based on human preadipocytes combined with gelatin microspheres containing basic fibroblast growth factor. *Biomaterials*. 2003;24:2513-2521.
24. Tamura E, Fukuda H, Tabata Y. Adipose tissue formation in response to basic growth factor. *Acta Oto-Laryngologica*. 2007;127:1327-1331.
25. Isogai N, Morotomi T, Hayakawa S, et al. Combined chondrocyte-copolymer implantation with slow release of basic fibroblast growth factor for tissue engineering an auricular cartilage construct. *J Biomed Mater Res A*. 2005;74:408-418.

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26. Iwakura A, Tabata Y, Tamura N, et al. Gelatin sheet incorporating basic fibroblast growth factor enhances healing of devascularized sternum in diabetic rats. *Circulation*. 2001;104:325-329.
27. Arnold GE. Vocal rehabilitation of paralytic dysphonia VIII. Phoniatric methods of vocal compensation. *Arch Otolaryngol*. 1962;76:76-83.
28. O'Leary MA, Grillone GA. Injection laryngoplasty. *Otolaryngol Clin N Am*. 2006;39:43-54.
29. Wang Y, Orbay H, Huang C, et al. Preclinical efficacy of slow-release bFGF in ischemia-reperfusion injury in a dorsal island skin flap model. *J Reconstr Microsurg*. 2013;29:341-346.
30. Shimizu A, Tajima S, Tobita M, et al. Effect of control-released basic fibroblast growth factor incorporated in β-Tricalcium phosphate for murine cranial model. *Plast Reconstr Surg Glob Open*. 2014;2:1-7.

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